Progressive Water Tariff Analysis

1. Problem Statement

Develop a Python program to calculate water bills for multiple households using consumption slabs, fixed charges, and penalties. Compare water consumption and identify households with excessive usage.

2. Objectives

1. Calculate water bills.
2. Apply different consumption slabs.
3. Add fixed charges and penalties.
4. Compare household consumption.
5. Identify excessive water usage.

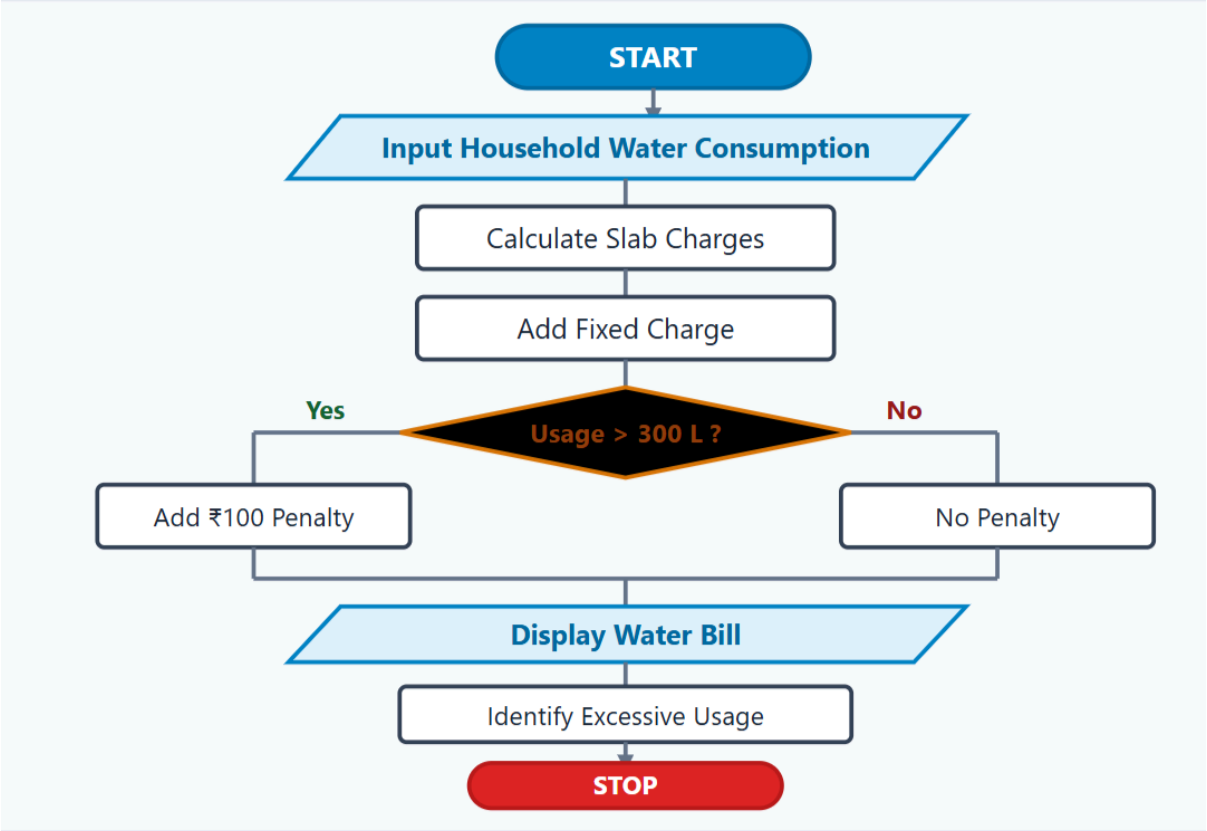
3. Assumptions

- First 100 litres: ₹2 per litre.
- Next 100 litres: ₹3 per litre.
- Above 200 litres: ₹5 per litre.
- Fixed charge: ₹50.
- Usage above 300 litres attracts a ₹100 penalty.
- Excessive usage means consumption above 300 litres.

4. Algorithm

1. Start.
2. Store household names and water consumption.
3. Initialize fixed charge and slab rates.
4. Read each household's consumption.
5. Calculate charges according to consumption slabs.
6. Add fixed charges.
7. Check whether consumption exceeds 300 litres.
8. Add penalty if required.
9. Display each household's bill.
10. Compare consumption of all households.
11. Identify excessive water usage.
12. Stop.

5. Flowchart



None

6. Pseudocode

None

```
START

Store household consumption

FOR each household:
    IF consumption <= 100:
        Calculate first slab charge
    ELSE IF consumption <= 200:
        Calculate first and second slab charges
    ELSE:
        Calculate all slab charges
```

```
    Add fixed charge

    IF consumption > 300:
        Add penalty

    Display bill

Identify highest consumption

STOP
```

8. Sample Output

```
None
House A Usage: 80 Litres, Bill: Rs. 210 Normal Usage
House B Usage: 150 Litres, Bill: Rs. 400 Normal Usage
House C Usage: 250 Litres, Bill: Rs. 900 Normal Usage
House D Usage: 400 Litres, Bill: Rs. 1750 Excessive Usage

Highest Consumption: House D
```

9. Test Cases

Test	Consumption	Expected Result
1	80 L	₹210
2	150 L	₹400
3	250 L	₹900

4	400 L	₹1750 + penalty included
5	350 L	Excessive usage

10. Conclusion

The Progressive Water Tariff Analysis system calculates household water bills using consumption slabs, fixed charges, and penalties. It identifies excessive usage and compares household consumption using Python.